

Dynamic Planet B/C - UT Invitational 2021 - **ANSWER KEY**

Section 1

Note for grading multiple-select questions: Correct answers earn 1 pt, incorrectly marked answers deduct 1 pt, and unmarked correct answers do not earn nor deduct points.

- | | | |
|-------------------|-----------------------|-----------------------|
| 1. C | 13. C | 25. D |
| 2. C | 14. A | 26. B, C |
| 3. D | 15. A, B, D | 27. D |
| 4. B | 16. A, B [TB5] | 28. C, E [TB3] |
| 5. A [TB4] | 17. B, C [TB6] | 29. A, B, D |
| 6. E | 18. B | 30. B |
| 7. D | 19. C | 31. C |
| 8. B | 20. A | 32. A |
| 9. B | 21. D | 33. A, B |
| 10. D | 22. B | 34. D [TB7] |
| 11. A | 23. A | 35. B, F |
| 12. C | 24. C | |

Section 2

36. a. Trellis b. A c. A

Note: Award 1 pt per row only if the entire row is correct

Question	Region 1	Region 2	Region 3	No region
37				✓
38				✓
39			✓	
40	✓	✓	✓	

41. B

42. **[TB8]**

Answer: 395 m/s^3

+1 for defining the flow rate as the product of the cross-sectional area and the velocity ($Q = Av$) (if this is implied and not directly stated, the point can still be earned; e.g., the student multiplies the area by 5)

+1 for correctly computing the cross-sectional area (79 m^2) (acceptable range: 77-81 m^2)

+1 for obtaining the correct answer with correct units (acceptable range: 385-405 m/s^3)

43. left side of stream, well sorted (+0.5 for each part correct)

44. Increase, because there would be more erosion, causing a deeper stream channel via downcutting. (+1 for "increase," +1 for a valid justification, such as increased erosion, deeper stream channel, and/or downcutting)

45. Downstream (+1), because the contour lines point in the upstream direction or opposite the direction of flow (+1)

46. 51-59 feet (Accept any whole number from 51 to 59 feet)

47. The dashed lines/hachure marks indicate a depression/zone of depression/area of elevation lower than the surroundings (response must mention "depression," "zone of depression," or indicate the meaning of the marks to receive credit)

48. Divide the sum of precipitation by the sum of evapotranspiration (or obtain the P/E_p ratio) (+2). Partial credit (+1) if response mentions summing the P and E_p for the entire year individually but does not mention division. -1 for each extraneous or incorrect calculation (e.g., subtraction, adding incorrect values).

49. November

50. February

51. E

52. **[TB2]** x - (fluvial) terrace
 y - (natural) levee
 z - yazoo tributary (Accept "yazoo," "yazoo stream," "yazoo channel")
 (Do NOT accept "tributary")

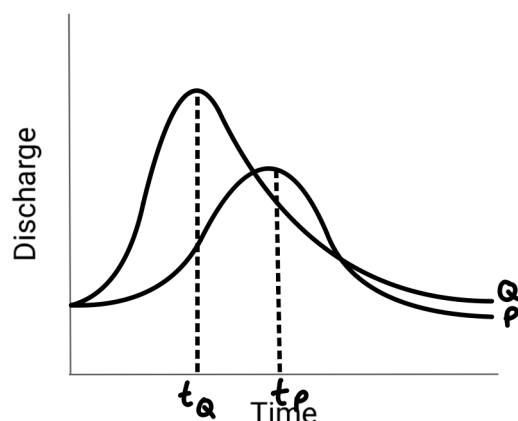
53. y, z (no partial credit)

54. a. 0.5% b. A, B (no partial credit)

55. D

56. a. E

b. Sample sketch:



+1: higher peak for curve Q and +1: sooner peak for curve Q

To earn both points, the graphs must have a generally correct shape (e.g., they resemble a sinusoidal curve rather than a parabolic one)

-1: no labels on x-axis (but graphs are labeled otherwise). No credit given if no labels exist at all on the answer sheet.

57. pressure, pressure difference/gradient, pressure gradient force (answer must mention pressure)

58. A, D (+1 if both, and only these letters, are given) because artesian wells are sourced from confined aquifers or because the wells are sourced from an aquifer enclosed between two impermeable layers / confining layers / aquicludes / shale layers. (+1 for valid justification - depth alone is NOT a valid justification)

59. Perched aquifer

60. **[TB1]** Award 1 point per row

Well	Level of the water table	Between water table and ground	Surface of the ground	Above ground
B	✓			
C	✓			
D		✓		

61. A

62. x, z (+1 for each letter, -1 for each extra letter, at least 0 pts)

63. a. B b. B

64. spring, autumn/fall

65. B

66. **[TB10]** a. seiche waves b. C, E (+1 for C or E, -1 for other letter, at least 0 pts)

67. **[TB9]** Negative feedback loop (+1), because

- a higher (lower) soil moisture content increases (decreases) evapotranspiration, which acts to reduce the gain (loss) in soil moisture
 - an increase (decrease) in evapotranspiration reduces (increases) soil moisture content, which reverses the initial increase (decrease) in evapotranspiration
- (+1 for a valid line of reasoning. Answers should correspond to one of the possible justifications above)

68. a. Poorly sorted, lowers hydraulic conductivity (+0.5 each)

b. Bog and marsh density is greater in areas affected by glaciation than areas not affected (the reverse is also true)

c. Low-order dominated

d. More tributaries; more well-drained (+0.5 each)